

THE RULES OF COMMUNICATION (PART 1)

3.1.2 COMMUNICATIONS FUNDAMENTALS

For communication to occur, every method has **three basic elements**:



1. Message Source (Sender)

The sender is a person or an electronic device that creates and sends the message.



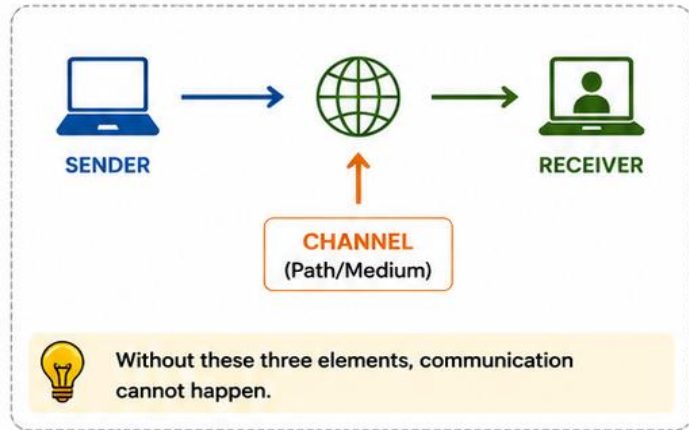
2. Message Destination (Receiver)

The receiver gets the message and understands (interprets) it.



3. Channel

The medium or path through which the message travels from sender to receiver.



3.1.3 COMMUNICATION PROTOCOLS

When sending a message (face-to-face or over a network), we follow certain set of rules called protocols.

These rules are specific to the communication method we use.

The process is similar to sending a letter.

EXAMPLE: SENDING A LETTER



Write the letter



Put in an envelope



Drop in mailbox



Delivered by post service



Receiver gets and reads it

Similarly, in computer networks, protocols define how messages are transmitted from sender to receiver.

3.1.4 RULE ESTABLISHMENT

Before communication, people must agree on rules (protocols) to govern the conversation.

Without Rules

humans communication between govern rules.
It is verydifficult tounderstand messages
that are not correctly formatted



With Rules

Rules govern communication between humans.
It is very difficult to understand
messages that are not correctly formatted.



Proper rules make the message clear, readable, and easy to understand.

Protocols must ensure the following for successful communication:



An identified sender and receiver



Common language and grammar



Speed and timing of delivery



Confirmation or acknowledgment requirements

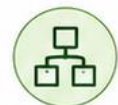
3.1.5 NETWORK PROTOCOL REQUIREMENTS

Network protocols not only identify the source and destination devices, but also define **HOW** a message is sent across a network.

Common requirements of computer network protocols:



1. Message Encoding



2. Message Formatting & Encapsulation



3. Message Size



4. Message Timing



5. Message Delivery Options

★ KEY TAKEAWAY

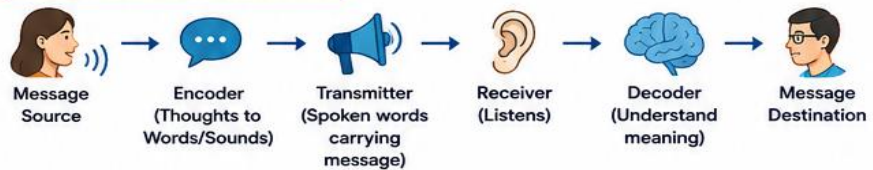
Communication works when there are senders, receivers, a channel, and agreed rules (protocols). Protocols ensure messages are sent correctly and understood by the receiver.

THE RULES OF COMMUNICATION (PART 2)

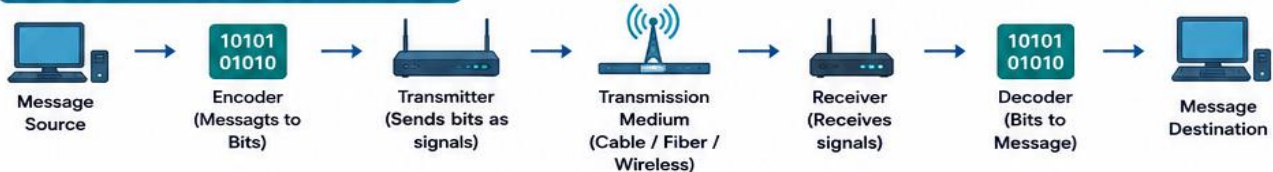
3.1.6 MESSAGE ENCODING

- **Encoding:** Converting information into another acceptable form for transmission.
- **Decoding:** Reverses the process to interpret the information.

ANALOGY (Human Communication)

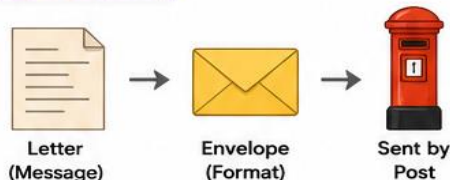


NETWORK EXAMPLE (Computer Communication)



3.1.7 MESSAGE FORMATTING AND ENCAPSULATION

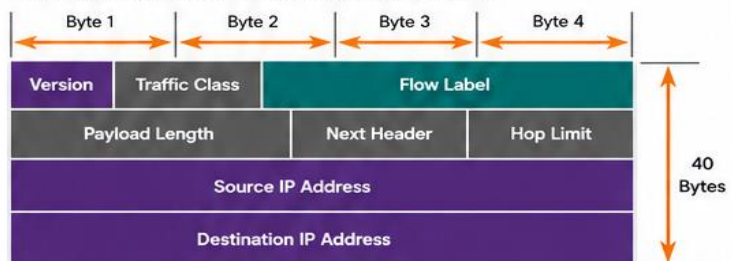
ANALOGY (Letter)



Placing one format (letter) inside another format (envelope) is called **ENCAPSULATION**. Decapsulation is the reverse process.

NETWORK EXAMPLE (IP Packet)

Data is sent in packets using specific formats. IPv6 packet fields identify source and destination.



IP is responsible for sending a message from source to destination over one or more networks.

3.1.9 MESSAGE TIMING



Flow Control

Manages the rate of data transmission. Ensures the sender does not send too much data too fast for the receiver.



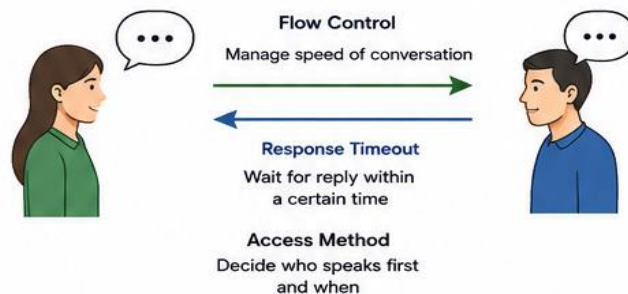
Response Timeout

If no response is received within a set time, the sender assumes failure and takes appropriate action (e.g., resend).



Access Method

Determines when a device can send data. Example: In WLAN, the NIC checks if the wireless medium is available before sending.



3.1.10 MESSAGE DELIVERY OPTIONS

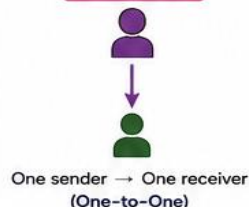
Messages can be delivered in different ways depending on the intended recipients.



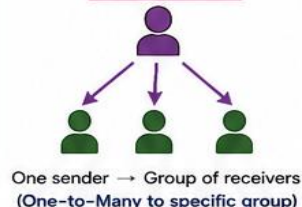
KEY TAKEAWAY

Rules like encoding, formatting, timing, and delivery ensure that messages are sent correctly and are understood by the intended receiver(s).

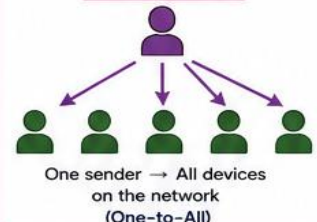
UNICAST



MULTICAST



BROADCAST



REMEMBER:

Right format + Correct timing + Proper delivery = Successful Communication

NETWORK PROTOCOLS – OVERVIEW & FUNCTIONS

3.2.1 NETWORK PROTOCOL OVERVIEW

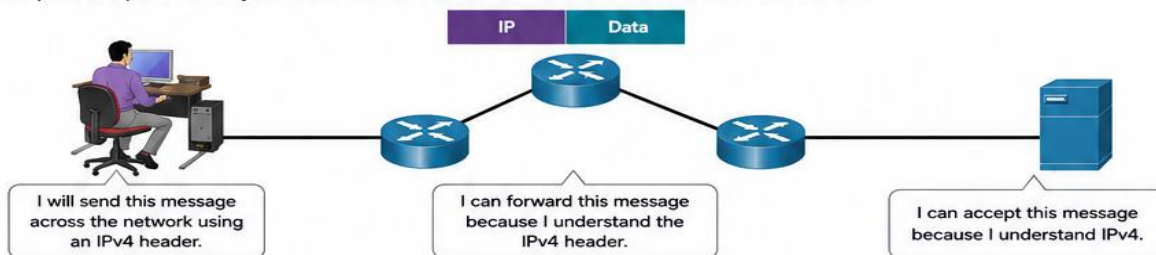


- For devices to communicate over a network, they must follow the same set of rules. These rules are called **PROTOCOLS**.
- Protocols define a common format and rules for exchanging messages between devices.
- Protocols are implemented by end devices and intermediary devices in software, hardware, or both.
- Different types of protocols are required to enable communications across one or more networks.

PROTOCOL TYPE	DESCRIPTION
 Network Communications Protocols	Enable two or more devices to communicate over one or more networks. The Ethernet family includes protocols such as IP, TCP, HTTP, and many more.
 Network Security Protocols	Secure data to provide authentication, data integrity, and data encryption. Examples: SSH, SSL, TLS.
 Routing Protocols	Enable routers to exchange route information, compare path information, and select the best path to the destination network. Examples: OSPF, BGP.
 Service Discovery Protocols	Used for automatic detection of devices or services. Examples: DHCP (discovers IP address), DNS (translates name to IP address).

3.2.2 NETWORK PROTOCOL FUNCTIONS

Network protocols perform many functions to ensure successful communication between end devices.

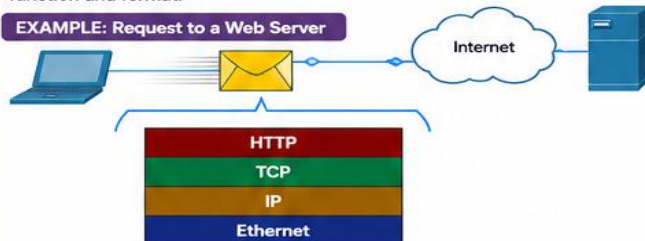


Computers and network devices use agreed-upon protocols to communicate.

FUNCTION	DESCRIPTION
 Addressing	Identifies the sender and intended receiver using a defined addressing scheme. Protocols that provide addressing: Ethernet, IPv4, IPv6.
 Reliability	Provides guaranteed delivery mechanisms in case messages are lost or corrupted in transit. TCP provides guaranteed delivery.
 Flow Control	Ensures that data flows at an efficient rate between two communicating devices. TCP provides flow control services.
 Sequencing	Labels each transmitted segment of data. The receiving device uses sequence numbers to reassemble information correctly. Useful if data is lost, delayed or received out-of-order. TCP provides sequencing services.
 Error Detection	Detects if data became corrupted during transmission. Examples of protocols that provide error detection: Ethernet, IPv4, IPv6, TCP.

3.2.3 PROTOCOL INTERACTION

A message sent over a network typically uses several protocols, each with its own function and format.



COMMON PROTOCOLS AND THEIR ROLES

- HTTP (HyperText Transfer Protocol)**
Defines the content and formatting of the request and response between client and web server (application layer).
- TCP (Transmission Control Protocol)**
Manages the conversation between end devices. Ensures reliable delivery and flow control.
- IP (Internet Protocol)**
Responsible for delivering packets from source to destination across multiple networks (routing).
- Ethernet**
Delivers frames from one device (NIC) to another on the same local area network (LAN).

KEY TAKEAWAY



- Protocols = Rules that govern communication in networks.
- They define formats, functions, and rules.
- Functions ensure messages are delivered correctly, securely, and efficiently.
- Multiple protocols work together in layers to complete communication.

Remember the Flow:

Application (HTTP)
↓
Transport (TCP)
↓
Internet (IP)
↓
Network Access (Ethernet)



NETWORK PROTOCOL SUITES (PART 1)

3.3.1 NETWORK PROTOCOL SUITES



For end devices to communicate over a network, they must follow the same set of rules called **PROTOCOLS**.



A **protocol suite** is a group of inter-related protocols that work together to perform a communication function.

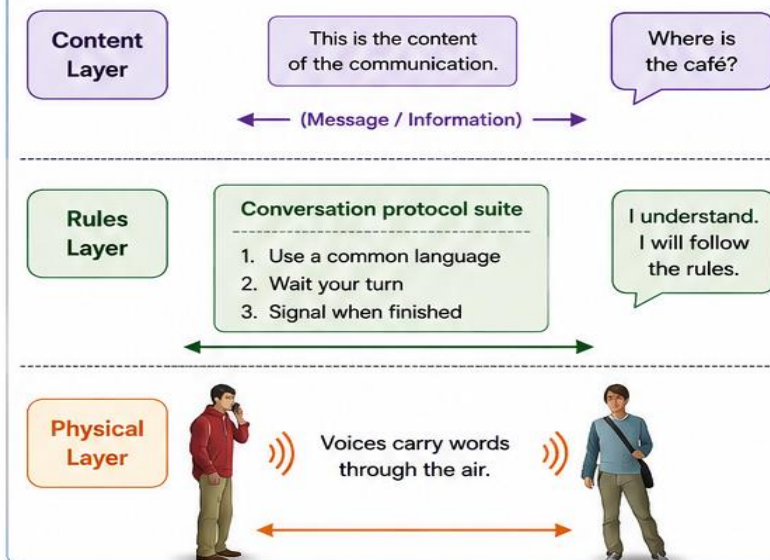


Protocol suites interact as a **stack of layers**, where each layer provides services to the layer above it.



The lower layers handle data movement and delivery across the network. The upper layers focus on the content of the message.

PROTOCOL SUITE AS A STACK – FACE-TO-FACE EXAMPLE



Protocol suites are sets of rules that work together to help solve a problem.

3.3.2 EVOLUTION OF PROTOCOL SUITES

- Since the 1970s, several protocol suites have been developed by standards organizations and vendors.
- Different protocol suites exist to provide comprehensive network communication services.

TCP/IP LAYER NAME	TCP/IP (SUITE)	ISO (SUITE)	AppleTalk (SUITE)	Novell NetWare (SUITE)
APPLICATION	HTTP DNS DHCP FTP	ACSE ROSE TRSE SESE	AFP	NDS
TRANSPORT	TCP UDP	TP0 TP1 TP2 TP3 TP4	ATP AEP NBP RTMP	SPX
INTERNET	IPv4 ICMPv4 IPv6 ICMPv6	CONP/CMNS CLNP/CLNS	AARP	IPX
NETWORK ACCESS	Ethernet ARP WLAN			

- **Internet Protocol Suite (TCP/IP)** – The most common and relevant protocol suite used today. It is an open standard maintained by the Internet Engineering Task Force (IETF).
- **Open Systems Interconnection (OSI) protocols** – A family of protocols developed jointly in 1977 by ISO and ITU. The OSI protocol also includes a seven-layer model called the OSI reference model.
- **AppleTalk** – A short-lived proprietary protocol suite released by Apple Inc. in 1985 for Apple devices. In 1995, Apple adopted TCP/IP to replace AppleTalk.
- **Novell NetWare** – A short-lived proprietary protocol suite and network operating system developed by Novell Inc. in 1983 using the IPX network protocol. In 1995, Novell adopted TCP/IP to replace IPX.



KEY TAKEAWAY



- ✓ A protocol suite = a group of protocols that work together.
- ✓ Protocols in a suite are organized in layers (stack).
- ✓ Lower layers: handle data movement & delivery across the network.
- ✓ Upper layers: focus on message content and user services.

NETWORK PROTOCOL SUITES (PART 2)

3.3.3 TCP/IP PROTOCOL EXAMPLE



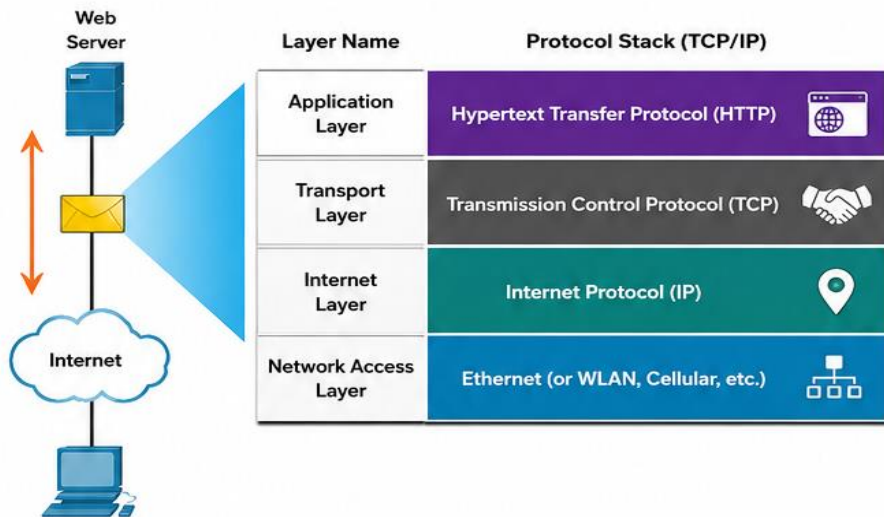
KEY POINTS

- TCP/IP protocols are used in the application, transport, and internet layers.
- There are no TCP/IP protocols in the network access layer.
- The most common network access layer LAN protocols are Ethernet and WLAN.
- TCP/IP protocols deliver the IP packet over the physical medium.

THREE TCP/IP PROTOCOLS USED

- ✓ **HTTP** – Application layer (Web communication)
- ✓ **TCP** – Transport layer (Reliable delivery)
- ✓ **IP** – Internet layer (Packet delivery across networks)

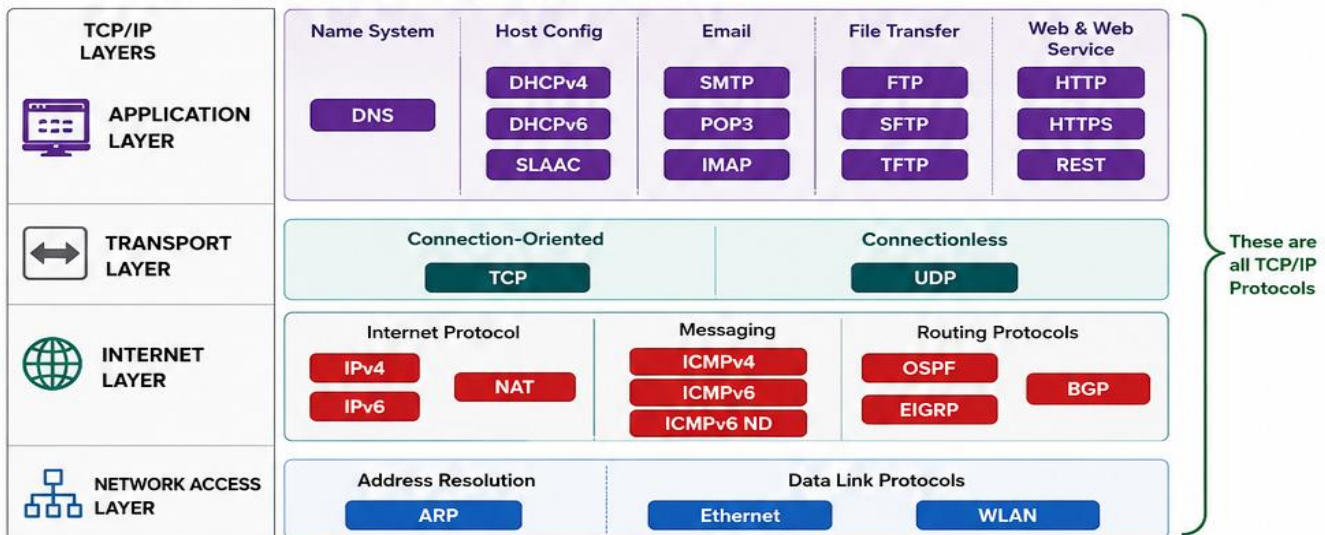
The figure shows how the three TCP/IP protocols work together to send packets between the web browser (host) and the web server.



HTTP creates the request/response, TCP ensures reliable delivery, IP delivers the packet to the destination across the network.

3.3.4 TCP/IP PROTOCOL SUITE

The TCP/IP suite includes many protocols and continues to evolve to support new services. Below are some of the important protocols organized by TCP/IP layers.



TWO IMPORTANT ASPECTS OF TCP/IP



Open Standard Protocol Suite
Freely available to the public.
Can be used by any vendor on hardware or in software.



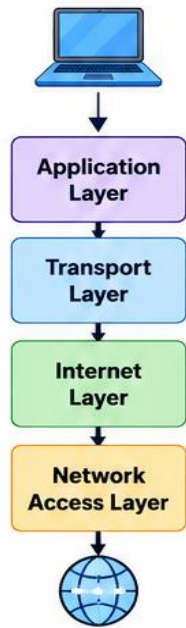
Standards-Based Protocol Suite
Endorsed by the networking industry and approved by a standards organization.
Ensures products from different manufacturers can interoperate successfully.



KEY TAKEAWAY

- ✓ Protocol suites group related protocols and work together to perform communication functions.
- ✓ TCP/IP is the most widely used suite in modern networks.
- ✓ Application (HTTP) + Transport (TCP) + Internet (IP) + Network Access (Ethernet/WLAN) = Complete Communication.

TCP/IP PROTOCOL SUITES – LAYER WISE PART 3



TCP/IP MODEL (4 Layers)

- ✓ **Application Layer**
– Provides network services to user applications.
- ✓ **Transport Layer**
– Ensures end-to-end communication between applications.
- ✓ **Internet Layer**
– Handles packet delivery across networks.
- ✓ **Network Access Layer**
– Transfers data over the physical network.

1 APPLICATION LAYER – Provides network services to user applications

 Name System	• DNS – Domain Name System. Translates domain names (e.g., cisco.com) into IP addresses.
 Host Config (Addressing)	• DHCPv4 – For IPv4. Dynamically assigns IPv4 addresses to clients at start-up and allows re-use when no longer needed. • DHCPv6 – For IPv6. Similar to DHCPv4, assigns IPv6 addresses to clients. • SLAAC – Stateless Address Autoconfiguration. Device obtains IPv6 address without using a DHCPv6 server.
 Email	• SMTP – Sends email from client to mail server and between servers. • POP3 – Retrieves email from mail server and downloads to client. • IMAP – Accesses email on server and maintains it there.
 File Transfer	• FTP – Access & transfer files to/from another host (reliable, connection-oriented). • SFTP – Secure FTP using SSH. File transfer is encrypted. • TFTP – Simple, connectionless file transfer with best-effort delivery.
 Web & Web Service	• HTTP – Transfers web content (text, images, audio, video, etc.). • HTTPS – Secure version of HTTP (encrypted). • REST – Web service using APIs and HTTP requests.

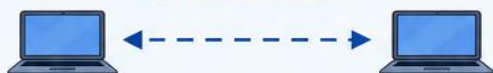
2 TRANSPORT LAYER – End-to-end Communication

Connection-Oriented



- ✓ **TCP – Transmission Control Protocol**
 - Provides reliable, acknowledged delivery.
 - Ensures data is delivered in order.
 - Used by most important applications (Web, Email, File Transfer).

Connectionless



- ✓ **UDP – User Datagram Protocol**
 - Sends packets without establishing connection.
 - Does not guarantee delivery (best-effort).
 - Faster, used where speed matters (e.g., streaming, DNS).

3 INTERNET LAYER – Packet Delivery Across Networks



- **IPv4** – Version 4. Uses 32-bit address. Receives segments from transport layer, packages into packets, and delivers end-to-end.
- **IPv6** – Version 6. Uses 128-bit address (larger addressing space).
- **NAT** – Network Address Translation. Converts private IPv4 addresses to public IPv4 addresses.



- **ICMPv4** – Provides feedback about errors in packet delivery from destination to source.
- **ICMPv6** – Similar to ICMPv4 but used for IPv6 packets.
- **ICMPv6 ND** – Neighbor Discovery. Used for address resolution and duplicate address detection in IPv6.



- **OSPF** – Open Shortest Path First. Link-state routing protocol based on areas (open standard).
- **EIGRP** – Enhanced Interior Gateway Routing Protocol (Cisco). Uses bandwidth, delay, load, reliability.
- **BGP** – Border Gateway Protocol (Exterior Gateway Protocol). Used between ISPs to exchange routing information.

4 NETWORK ACCESS LAYER – Access to Physical Network

Address Resolution



- **ARP** – Address Resolution Protocol. Maps IPv4 address → MAC address. Used at Internet Layer (OSI L3).



Note: You may see ARP at L3 in some documentation, but in TCP/IP model it is part of Network Access Layer.

Data Link Protocols (How data is sent on the local network)



Ethernet – Defines rules for writing and signaling standards of the network access layer. Used in wired LANs.



WLAN (Wireless LAN) – Defines rules for wireless transmission over a LAN. Used in Wi-Fi networks.

QUICK RECAP TABLE

	Main Role
Application	Provides network services to user applications (Web, Email, File, etc.)
Transport	End-to-end communication (TCP reliable / UDP best-effort)
Internet	Logical addressing and routing (IP, ICMP, OSPF, BGP, etc.)
Network Access	Access to physical network (Ethernet, WLAN, ARP, etc.)

KEY TAKEAWAYS

- ✓ Each layer has specific protocols and functions.
- ✓ Together, these layers provide complete communication from your device to any server.
- ✓ TCP/IP suite is the most widely used suite today.



STANDARDS ORGANIZATIONS

3.4.1 OPEN STANDARDS



Open standards ensure interoperability, competition, and innovation. They allow products from different vendors to work together (interconnect).



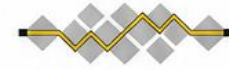
KEY POINTS

- Many products from different vendors must use the same standards to work together.
- Open standards are developed by standards organizations.
- Standards can be open (available to anyone) or proprietary (controlled by one vendor).
- Open standards are usually vendor-neutral and maintained by nonprofit organizations.
- They ensure that products from multiple vendors can interoperate on the same network.
- Example: Devices in your home network use open standards like IPv4, IPv6, DHCP, SLAAC, Ethernet, and 802.11 WLAN. These standards help your laptop, phone, and server run Linux or another OS communicate with each other.

MAJOR STANDARDS ORGANIZATIONS



IEEE (Institute of Electrical and Electronics Engineers)



IETF (Internet Engineering Task Force)



IANA (Internet Assigned Numbers Authority)



ICANN (Internet Corporation for Assigned Names and Numbers)



ITU (International Telecommunication Union)



TIA (Telecommunications Industry Association)



KEY TAKEAWAY

- ✓ Open standards = Interoperability + Innovation + Competition
- ✓ Used by both operating systems and applications
- ✓ Developed and maintained by standards organizations
- ✓ Example standards: IPv4, IPv6, DHCP, SLAAC, Ethernet, 802.11 WLAN

3.4.2 INTERNET STANDARDS



Several organizations work together to develop and support Internet and TCP/IP standards.

INTERNET STANDARDS ORGANIZATIONS



Internet Society (ISOC)
Leads the open development and evolution of the Internet globally.



Internet Architecture Board (IAB)
Provides architectural guidance for the Internet.



Internet Engineering Task Force (IETF)
Develops Internet and TCP/IP standards.

Internet Engineering Steering Group (IESG)

Working Group #1

Working Group #2



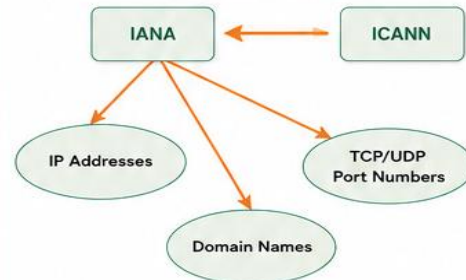
Internet Research Task Force (IRTF)
Focuses on long-term research and innovation.

Internet Research Steering Group (IRSG)

Research Group #1

Research Group #2

IANA & ICANN ROLES



IANA – Manages the technical information resources at the core of the Internet.

ICANN – Oversees IP address allocation, domain name management, and assignment of other information used in TCP/IP.

3.4.3 ELECTRONIC AND COMMUNICATIONS STANDARDS



Other organizations promote and create electronic and communication standards used to deliver electronic signals over wired or wireless media.

MAJOR ORGANIZATIONS AND THEIR FOCUS

- **IEEE (Institute of Electrical and Electronics Engineers)** – Advancing technology innovation and creating standards in many areas, including power, energy, electronics, telecommunications, and networking. Example: 802.3 Ethernet and 802.11 WLAN standards.
- **EIA (Electronic Industries Alliance)** – Sets standards for electrical wiring, connectors, and infrastructure used in networking equipment.
- **TIA (Telecommunications Industry Association)** – Develops standards for a wide range of areas including cables, towers, VoIP, wireless devices, satellites, and more.
- **ITU-T (International Telecommunication Union – Telecommunication Standardization Sector)** – Defines global standards for video, Internet Protocol Television (IPTV), and broadband communications (e.g., DSL).

WHAT DO THEY STANDARDIZE?



Electrical wiring and connectors



Cables, towers, and infrastructure



Voice over IP (VoIP), wireless devices, satellites



Video, IPTV, and broadband services

REFERENCE MODEL (PART 1)

3.5.1 THE BENEFITS OF USING A LAYERED MODEL



KEY IDEA

We cannot actually watch real packets travel across a real network.
A model helps us think about a network so we can imagine what is happening.

WHY A LAYERED MODEL?

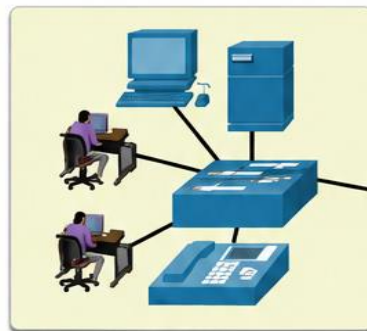
Network concepts can be complex and hard to understand. A layered model breaks the operations of a network into manageable layers so it is easier to explain and understand how a network operates.

BENEFITS OF USING A LAYERED MODEL

- ✓ **Assists in protocol design** – Each layer has a defined job and a clear interface with the layers above and below.
- ✓ **Fosters competition** – Different vendors can develop products that work together.
- ✓ **Prevents changes from spreading** – Changes in one layer do not affect other layers above or below.
- ✓ **Provides a common language** – Helps everyone describe networking functions and capabilities.

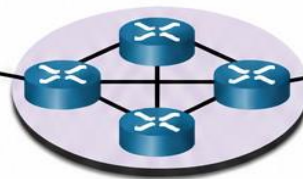
LAYERED MODELS USED

- Open Systems Interconnection (OSI) Reference Model
- TCP/IP Reference Model

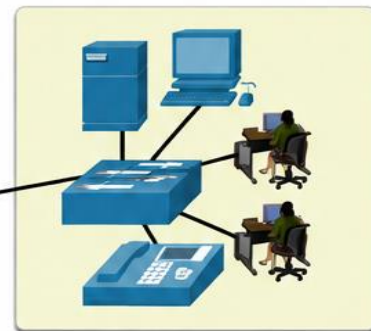


OSI Model

A networking model is only a representation of a network operation.
The model is not the actual network.



TCP/IP Protocol Suite



TCP/IP Model

Application	TCP/IP Protocol Suite	TCP/IP Model
Presentation	HTTP, DNS, DHCP, FTP	Application
Session		
Transport	TCP, UDP	Transport
Network		
Data Link	IPv4, IPv6, ICMPv4, ICMPv6	Internet
Physical	Ethernet, WLAN, SONET, SDH	Network Access



KEY TAKEAWAY

- ✓ A layered model helps us understand how networks operate.
- ✓ It organizes operations into layers with specific jobs.
- ✓ Two main models describe network operations: OSI and TCP/IP.
- ✓ These models are representations, not the actual network.



Reference Model (OSI and TCP/IP)



OSI Reference Model

- Provides a standard 7-layer model for network communication.
- Helps different vendors and systems work together.
- Works as a guideline for understanding how data moves across a network.

OSI Layers (7)

- 7 Application
- 6 Presentation
- 5 Session
- 4 Transport
- 3 Network
- 2 Data Link
- 1 Physical



TCP/IP Protocol Model

- Used in real Internet (early 1970s).
- Sometimes called the Internet model.
- Divides functions into 4 layers.

TCP/IP Layers (4)

- 4 Application
- 3 Transport
- 2 Internet
- 1 Network Access



OSI vs TCP/IP Model Comparison

Feature	OSI Model	TCP/IP Model
Total Layers	7 layers	4 layers
Main Idea	Reference model	Practical / Internet model
Use	Conceptual understanding	Real-world Internet



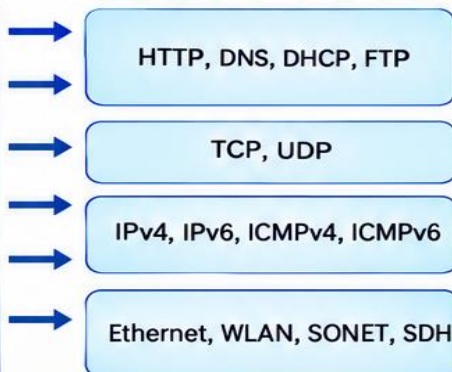
HTTP-based TCP/IP Protocol Suite

TCP/IP suite includes protocols for each layer:

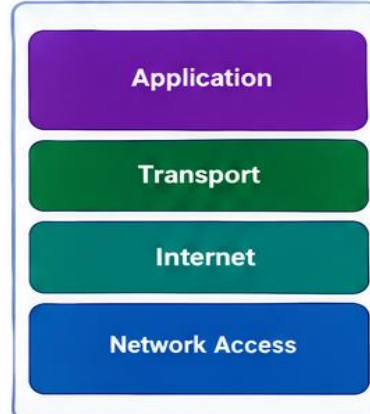
OSI Model (7 Layers)



Main Protocols Used



TCP/IP Model (4 Layers)



Key Takeaway: OSI has 7 layers (conceptual model) while TCP/IP has 4 layers (practical Internet model). TCP/IP is the foundation of today's Internet communication.



Reference Model – OSI & TCP/IP



OSI Reference Model (7 Layers)

- ✓ Open standard model for network communication.
- ✓ Divides communication into 7 layers.
- ✓ Each layer has specific functions and communicates with adjacent layers.

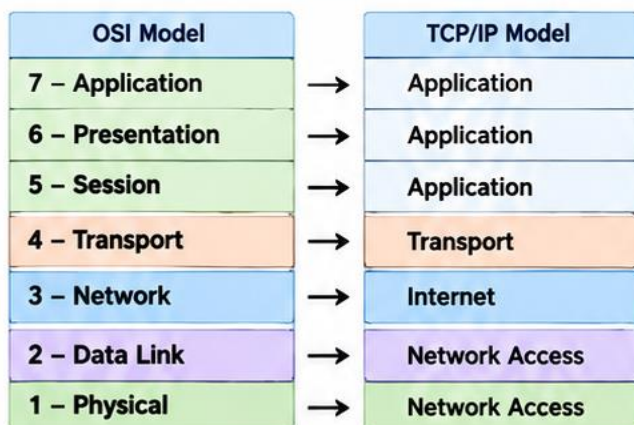


TCP/IP Model (4 Layers)

- ✓ Internet model created in the 1970s.
- ✓ Sometimes called the Internet model.
- ✓ Divides communication into 4 layers.
- ✓ TCP/IP is a reference model.



OSI vs TCP/IP Model



Layer Functions (TCP/IP)

- **Application:** Uses protocols for process-to-process communication.
- **Transport:** Supports communication between different devices across diverse networks.
- **Internet:** Determines the best path through the network.
- **Network Access:** Controls hardware devices and media.

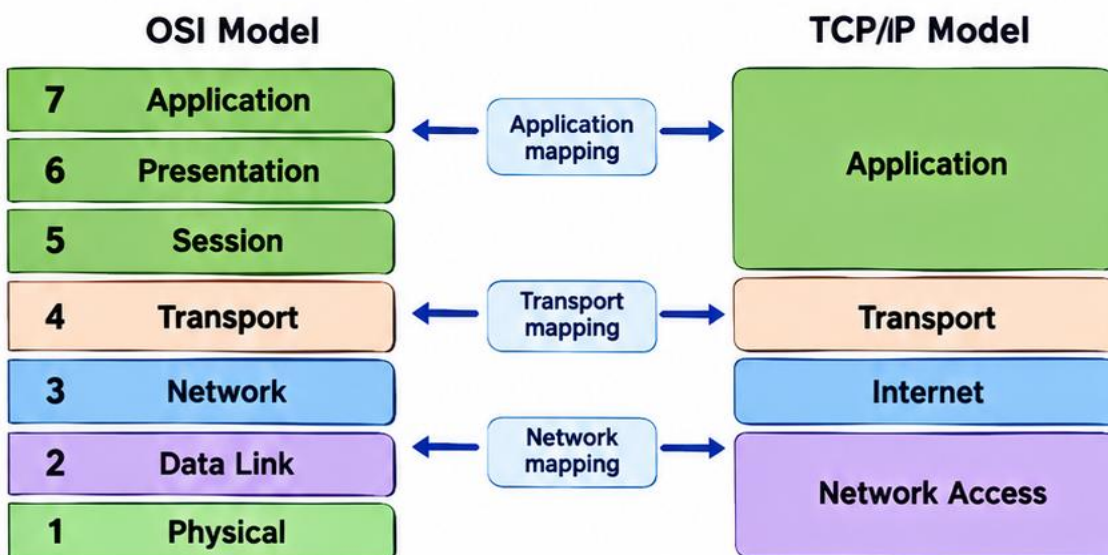


Key Point:

OSI has 7 layers while TCP/IP has 4 layers. TCP/IP layers are a practical implementation of the OSI model.



OSI and TCP/IP Layer Mapping



Remember: OSI – 7 layers (theory model) and TCP/IP – 4 layers (practical model).



DATA ENCAPSULATION (PART 1)

1 3.6.1 SEGMENTING MESSAGES



Why Segmentation?

- Large messages (videos, emails, files) are divided into smaller pieces so they can be sent across networks.
- If a network link fails, only the pieces being sent on that link may need to be resent – not the entire message.



Benefits

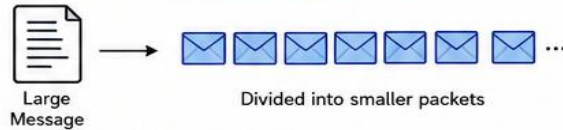
- **Increases speed** – Large data is broken into packets. Different conversations (multiplexing) can share the network link.
- **Increases efficiency** – If only one piece is lost, only that piece needs to be resent, not the whole message.



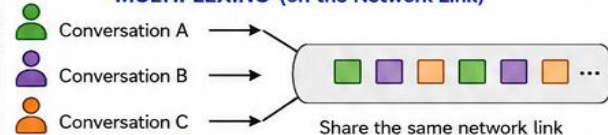
Example: A video file is divided into many small packets. Each packet can take a different route and reassemble at the destination.

Segmentation vs. Multiplexing

SEGMENTATION (at Sender)



MULTIPLEXING (on the Network Link)



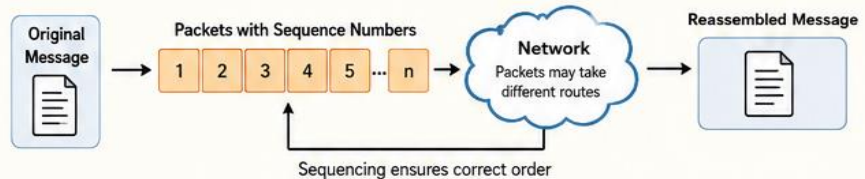
Key Takeaway

Segmentation makes data manageable and reliable by dividing it into packets.

2 3.6.2 SEQUENCING

Each packet is numbered (sequenced) before sending.

At the destination, packets are reassembled in the correct order to recreate the original message.



3 3.6.3 PROTOCOL DATA UNITS (PDUs)

As data travels down the protocol stack, each layer adds its own information (header/trailer). This process is called encapsulation.



Note:

- UDP PDU is called a datagram.
- IP packets are sometimes also called IP datagrams.

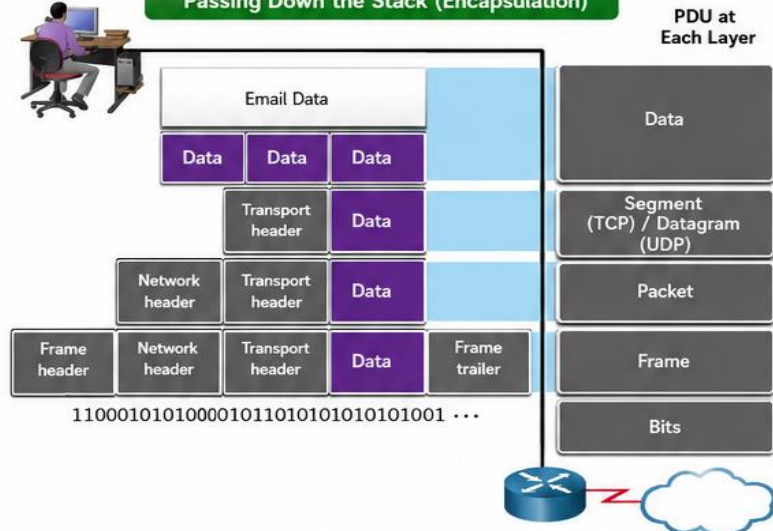
PDU Names by Layer

Layer	PDU Name
Application	Data
Transport	Segment (TCP) / Datagram (UDP)
Network (Internet)	Packet
Data Link	Frame
Physical	Bits



Note: If the Transport header is TCP → PDU = Segment.
If the Transport header is UDP → PDU = Datagram.

Passing Down the Stack (Encapsulation)



Summary of PDUs

- **Data** – General term for application layer PDU.
- **Segment** – Transport layer PDU (when using TCP).
- **Datagram** – Transport layer PDU (when using UDP).
- **Packet** – Network layer PDU.
- **Frame** – Data link layer PDU.
- **Bits** – Physical layer PDU (actual 0s and 1s transmitted on the medium).



DATA ENCAPSULATION (PART 2)

1 3.6.4 ENCAPSULATION EXAMPLE

► Encapsulation works from top to bottom (sender side).

- At each layer, the upper layer information is treated as **DATA** within that layer's protocol.
- Example: The TCP segment is treated as DATA inside the IP packet.



Key Idea

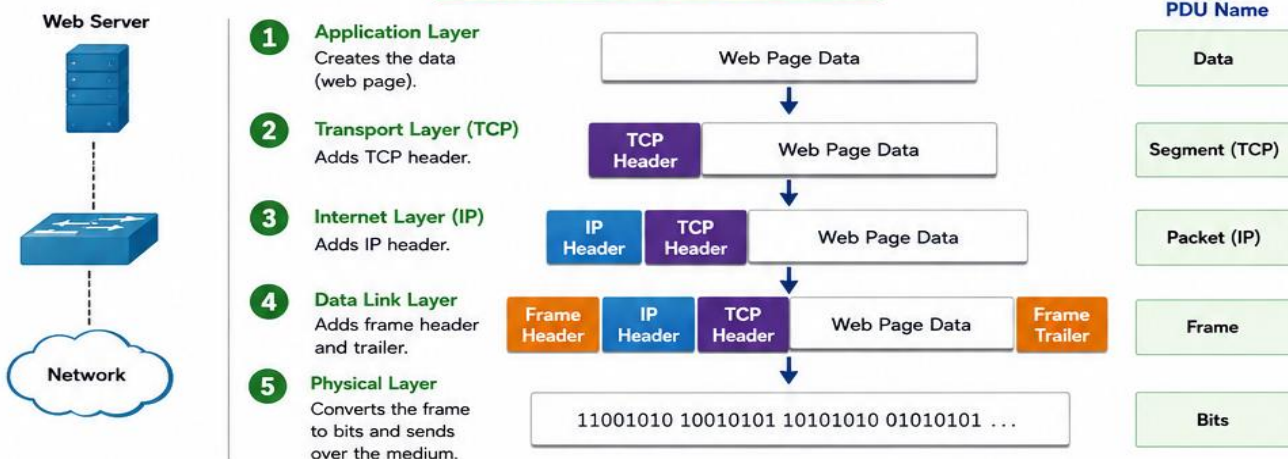
Each layer adds its own header (and sometimes a trailer) to prepare the data for transmission over the network.



Example Scenario

A web server sends a web page to a web client. The message is encapsulated as it moves down the protocol stack before being transmitted.

ENCAPSULATION PROCESS (Sender)



Result: The data is now ready to be transmitted across the network.

2 3.6.5 DE-ENCAPSULATION EXAMPLE

► De-encapsulation works from bottom to top (receiver side).

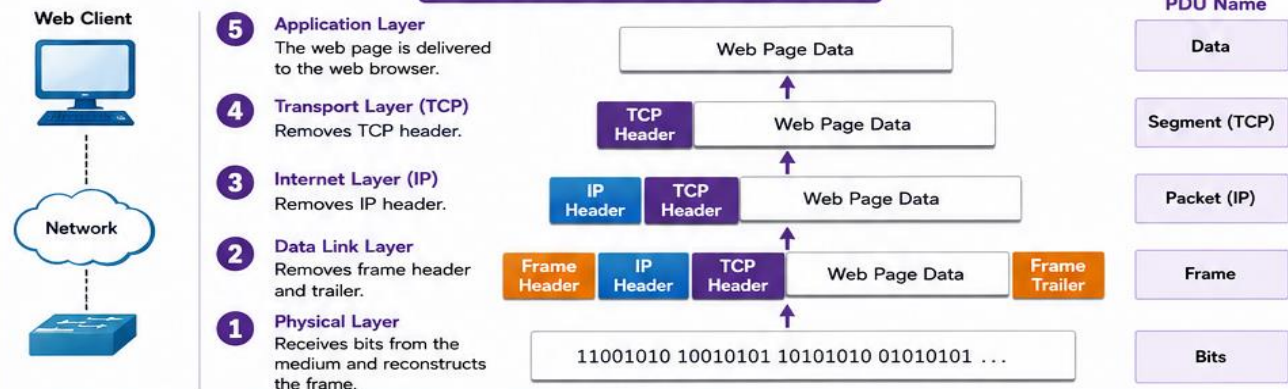
- The receiving device removes one or more protocol headers at each layer.
- The data is recovered and delivered up the stack to the end-user application.



Key Idea

Each layer removes its own header (and trailer) to extract the original data.

DE-ENCAPSULATION PROCESS (Receiver)



Result: The original web page is recovered and shown to the user.



DATA ACCESS (PART 1)

1 3.7.1 ADDRESSES



Why Addresses Are Needed

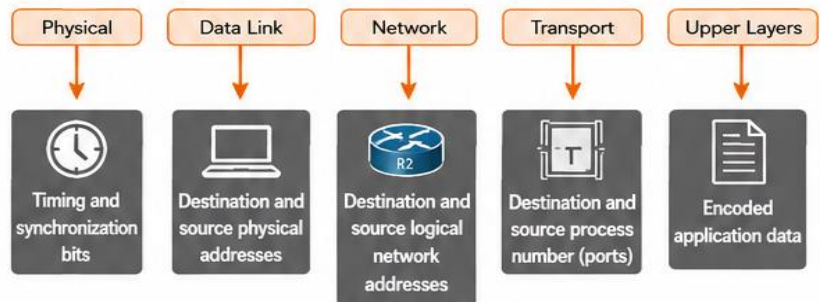
- Messages are segmented before sending.
- If not addressed properly, segments cannot be delivered to the right destination.
- This topic gives an overview of network addresses.



Layer Responsibilities

- **Network layer** – Delivers the IP packet from the original source to the final destination (same or remote network).
- **Data link layer** – Delivers the data link frame from one NIC to another NIC on the same network.

Addresses at Each Layer (from top to bottom)

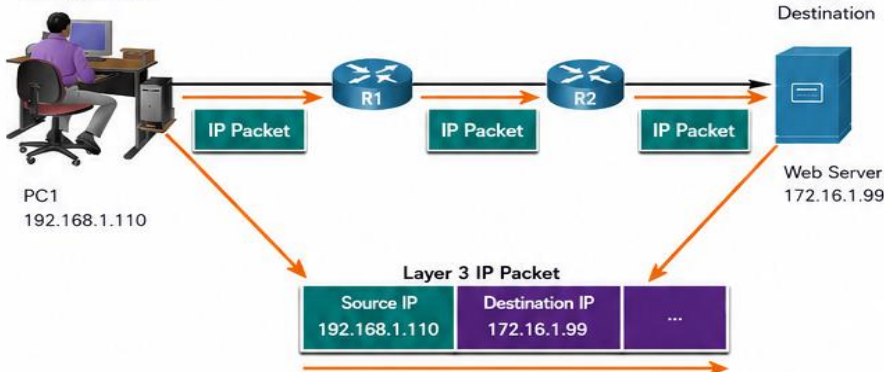


2 3.7.2 LAYER 3 LOGICAL ADDRESS

An IP address is the network layer (Layer 3) logical address used to deliver the IP packet from the original source to the final destination.

Original Source

Final Destination



What is in an IP Packet?

The IP packet contains two IP addresses:

- **Source IP address** – The IP address of the sending device (original source).
- **Destination IP address** – The IP address of the receiving device (final destination).

IP Address Structure

An IP address has two parts:

- **Network portion (IPv4) or Prefix (IPv6)** – Identifies the network to which the address belongs. Devices on the same network have the same network portion.
- **Host portion (IPv4) or Interface ID (IPv6)** – Identifies a specific device (unique to that device or interface) on the network.



Same Network or Different Network?

The IP addresses indicate the original source IP address and final destination IP address. This is true whether the source and destination are on the same IP network or different IP networks.



Note: The subnet mask (IPv4) or prefix-length (IPv6) is used to separate the network portion from the host portion.

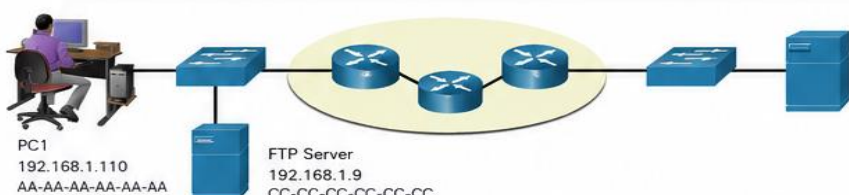
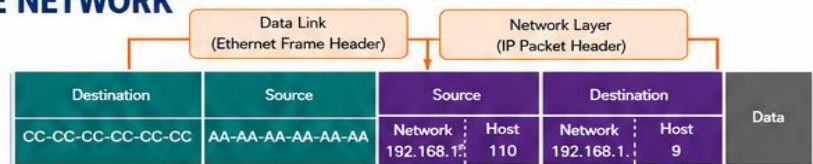
3 3.7.3 DEVICES ON THE SAME NETWORK

Example: A client computer (PC1) communicates with an FTP server on the same IP network.

- **Source IPv4 address** – The IPv4 address of the sending device, PC1: 192.168.1.110.
- **Destination IPv4 address** – The IPv4 address of the receiving device, FTP server: 192.168.1.9.



Notice: The network portion of the source IPv4 address and the network portion of the destination IPv4 address are the same. Therefore, the source and destination are on the same network.



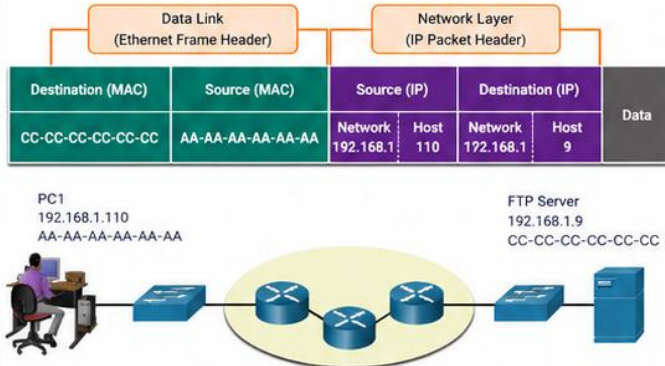


DATA ACCESS (PART 2)

4 3.7.4 ROLE OF THE DATA LINK LAYER ADDRESSES – SAME IP NETWORK



When sender and receiver are on the same IP network, the data link frame is sent directly to the receiving device.

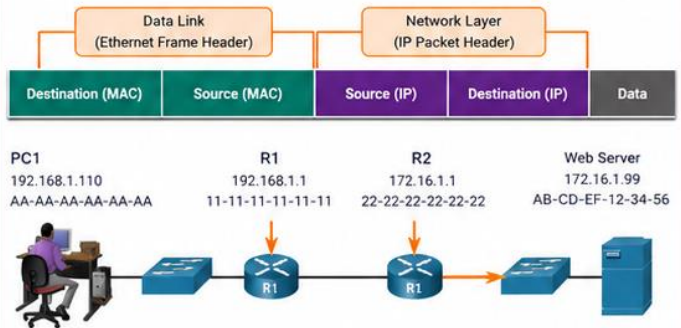


- **Source MAC address** – MAC of the sending device (PC1): AA-AA-AA-AA-AA-AA
- **Destination MAC address** – MAC of the receiving device (FTP server): CC-CC-CC-CC-CC-CC
- **Network portions of source and destination IP are the same (192.168.1) → Same network**
- Frame is sent directly from PC1 to FTP server.

5 3.7.5 DEVICES ON A REMOTE NETWORK



When devices are on different IP networks, packets must be routed through one or more routers to reach the final destination.



IP Addresses

- **Source IPv4 address (PC1):** 192.168.1.110
- **Destination IPv4 address (Web Server):** 172.16.1.99
- **Network portions are different (192.168.1 ≠ 172.16.1) → Remote network**

6 3.7.6 ROLE OF THE NETWORK LAYER ADDRESSES



When communicating with a device on a different network:

- **Source IP address** = IP of sending device (host)
- **Destination IP address** = IP of the final destination device (host on remote network)

The router uses the destination IP address to determine the best path to the remote network.



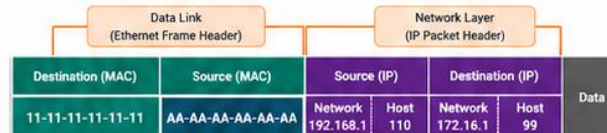
Key Point

IP addresses do not change from source to destination across the entire journey.

7 3.7.7 ROLE OF THE DATA LINK LAYER ADDRESSES – DIFFERENT IP NETWORKS



Frames cannot be sent directly to a remote host. Instead, frames are sent to the next hop router (or default gateway) on the same network.



Each hop (PC1 → R1, R1 → R2, R2 → Server) uses new data link (MAC) addresses for that hop only. But the IP addresses (source & destination) remain the same end-to-end.

What do the MAC addresses mean?

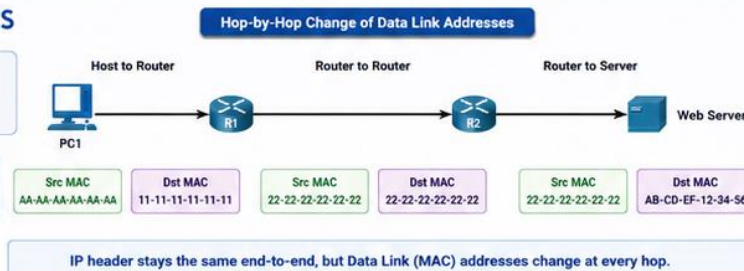
- **Source MAC address** – MAC of the sending device (PC1): AA-AA-AA-AA-AA-AA
- **Destination MAC address** – MAC of the next hop router (R1) interface on same network: 11-11-11-11-11-11

8 3.7.8 DATA LINK ADDRESSES

Layer 2 (Data Link) addresses have a different role: They deliver the data link frame from one network interface to another interface on the same network.



Before an IP packet is sent over a wired or wireless network, it must be encapsulated in a data link frame.



Key Takeaway

IP addresses = identify end hosts across different networks.

MAC addresses = identify interfaces on the same network (one hop only).